

Vectors → Matrices → Regression

Explore cosine similarity, dot products, matrix multiplication, projection shapes, and how all of these ideas link to linear regression.

1 Two Vectors

A vector can be represented as a **column of numbers**. Change the values below and watch the vectors move on the coordinate plot.



Cosine of the angle

$$\cos \theta = \mathbf{a} \cdot \mathbf{b} / (\|\mathbf{a}\| \|\mathbf{b}\|)$$

The dot product tells us how strongly the two vectors point in the same direction. Cosine similarity focuses only on their direction, not their size.

$$\cos \theta = 0.2333$$

$$\theta = 76.51^\circ$$

Dot product: multiply and sum

$$a_1b_1 + a_2b_2 + a_3b_3$$

$$-4 \times 1 + 2 \times 3 + 1 \times 2$$

$$\mathbf{a} \cdot \mathbf{b} = 4.0000$$

Each corresponding pair is multiplied, and then all those products are added together.

We usually assume $\mathbf{a}$ and $\mathbf{b}$ are **unit vectors** and therefore:

$$\cos \theta = \mathbf{a} \cdot \mathbf{b}$$

$$\text{cosine similarity} = \text{dot product}$$

Higher similarity $\rightarrow$ Higher dot product

Lower similarity $\rightarrow$ Lower dot product

$\cos \theta = 0$ degree $\rightarrow \cos(\theta) = 1$

$\cos \theta = 90$ degree $\rightarrow \cos(\theta) = 0$

$\cos \theta = 180$ degree $\rightarrow \cos(\theta) = -1$

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Why Transpose?

Shape in

orange

Both **a** and **b** are column vectors:

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \times \begin{matrix} \text{Not matrix multiplication compatible} \\ (3 \times 1)(3 \times 1) \end{matrix}$$

3×1 3×1

$$a^T \cdot b$$

$$\begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{matrix} \text{Scalar} \\ 1 \times 1 \end{matrix}$$

1×3 3×1

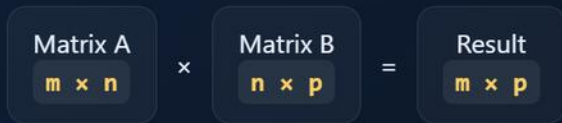
$$a^T b = a_1 b_1 + a_2 b_2 + a_3 b_3$$

4.0000

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Matrix Multiplication

The shape rule



The **inside dimensions must match**. The outside dimensions become the shape of the answer.

$$(AB)_{ij} = \sum a_{ik} b_{kj}$$

5 Matrix Multiplication — Step by Step

$$\begin{matrix} \text{A} \\ 4 \times 2 \end{matrix} \times \begin{matrix} \text{B} \\ 2 \times 3 \end{matrix} = \begin{matrix} \text{AB} \\ 4 \times 3 \end{matrix}$$

$$\begin{matrix} \text{A} \\ 4 \times 2 \end{matrix} \times \begin{matrix} \text{B} \\ 2 \times 3 \end{matrix} = \begin{matrix} \text{AB} \\ 4 \times 3 \end{matrix}$$

1	2
3	4
5	6
7	8

2	1	3
4	2	5

10	5	13
22	11	29
34	17	45
46	23	61

Next step →

Reset

Step 12: Row 4 of A × Column 3 of B

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What Shape Projects a Vector?

Suppose we have a vector with shape 2×1 . What matrix is required to transform it into a 3×1 vector?

$$2 \times 2$$

Can it produce 3×1 ?

Check

$$3 \times 2$$

Can it produce 3×1 ?

Check

$$2 \times 3$$

Can it produce 3×1 ?

Check

$$(3 \times 2)(2 \times 1) = 3 \times 1$$

Therefore, to map 2-dimensional $\rightarrow$ 3-dimensional, we need a matrix of shape 3×2 .

Now suppose we want to map a 2×1 vector into a 5×1 vector.

$$(5 \times 2)(2 \times 1) = 5 \times 1$$

Required projection matrix:

$$5 \times 2$$

7 From Linear Regression to Vectors

We normally begin with the equation of a straight line:

$$y = mx + c \rightarrow y = w_1x + w_0$$

$$y = w_1x + w_0 \quad (1)$$

We can make this look like a dot product by putting the weights and input values into vectors.

$$\begin{array}{c} \mathbf{w}^T \\ \left[\begin{array}{cc} w_1 & w_0 \end{array} \right] \\ 1 \times 2 \end{array} \cdot \begin{array}{c} \mathbf{x} \\ \left[\begin{array}{c} x \\ 1 \end{array} \right] \\ 2 \times 1 \end{array} = \begin{array}{c} y \\ 1 \times 1 \end{array}$$

$$y = [w_1 \ w_0] [x \ 1]^T$$

$$y = \mathbf{w}^T \mathbf{x}$$

The same idea:
multiply corresponding values → add them